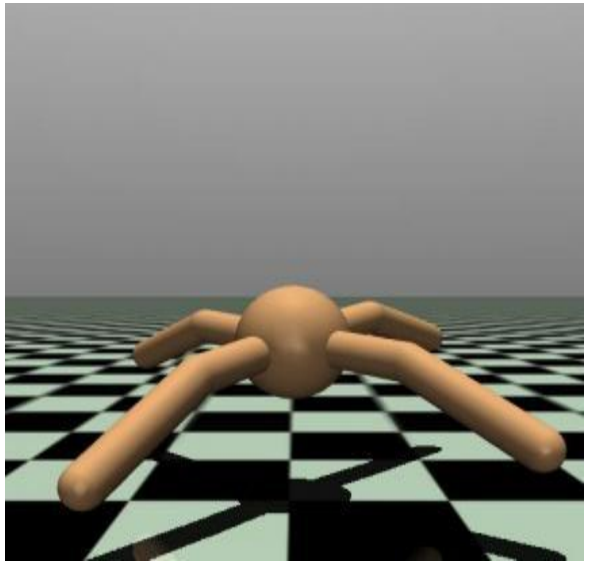
Autonomous Ant Robot Control via Deep Reinforcement Learning

Sole Author and Architect | Independent Researcher

Implemented a complete Deep RL pipeline (PPO) to control a multi-legged robot (Ant-v4) in the MuJoCo physics environment.

Trained a 500,000-step policy on a cloud NVIDIA RTX 4090 GPU, achieving robust locomotion.

Published on GitHub for public verification (seattle27/Ant_AI_Robot).



EDUCATION & PROFESSIONAL EXPERIENCE

Bachelor of Science (B.S.) in Electrical Engineering and Automation | Qingdao City University | Graduated in 2021

Completed a rigorous undergraduate curriculum, establishing strong theoretical foundations in control systems, circuit analysis, signal processing, and applied mathematics.

5+ Years of Independent Professional R&D Experience (2021 - Present)

Since graduation, has been continuously and independently engaged in advanced interdisciplinary research across Electrical Engineering, Computer Science, Mathematics, and Physics.

This extended professional experience, combined with the formal undergraduate degree in a STEM field, fully satisfies the "Advanced Degree" requirement for the EB-2 category.

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Extensive Academic Repository & Deep Theoretical Foundations

A massive, curated collection of graduate-level textbooks, top-tier conference papers (e.g., NeurIPS), advanced MIT OpenCourseWare lectures, and comprehensive study materials across Mathematics, Physics, and Computer Science. This repository serves as the primary evidence of an 8-year, uninterrupted, and highly rigorous self-directed education.

Advanced Graduate & Doctoral Level Coursework (Self-Studied)

Successfully completed the comprehensive core curriculum expected of a Ph.D. student in pure mathematics and theoretical physics, including:

Real Analysis & Functional Analysis (Walter Rudin)

Differential Geometry & Topology (Graduate Texts in Mathematics)

Differential Forms in Algebraic Topology (Raoul Bott)

Advanced Number Theory (Basic Algebraic Geometry, Algebraic Topology)

Classical & Quantum Mechanics (Physics core)

Advanced Computer Science (Data Structures, Algorithms, Operating Systems, ML Theory)